Hotel Booking Demand Analysis

Data cleaning, exploratory analysis and business insights using Python

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Programme: AltSchool Africa, Data Analytics (Third Semester Project)
Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook
Dataset: Hotel Booking Demand (City Hotel and Resort Hotel)

119,390	87,228	27.5%	59%
raw records	clean bookings	cancellation rate	of bookings from Online TA

1. The business problem

Hotels lose money when bookings are cancelled, and they leave money on the table when they misread demand. This project analyses two hotels (a City Hotel and a Resort Hotel) to answer: **where does demand come from, which bookings get cancelled, and what drives revenue?**

2. Data cleaning

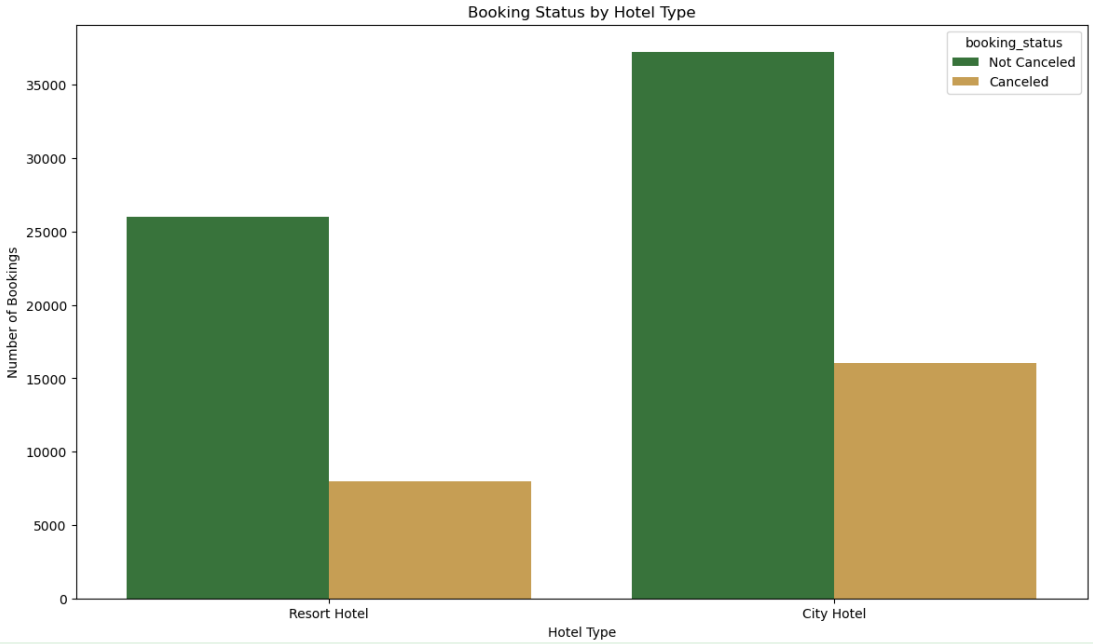
The raw file had 119,390 rows and 32 columns. Each issue was documented first, then fixed, then verified.

Issue found	Action taken	Result
31,994 exact duplicate rows (26.8%)	Removed	87,396 rows
children: 4 missing	Filled with median (0)	0 missing
country: 452 missing	Filled with 'Unknown'	0 missing
agent: 12,193 missing	Filled with 'Unknown'	0 missing
company: 82,137 missing	Filled with 'Unknown'	0 missing
166 bookings with zero guests	Removed (no adults, children or babies)	87,230 rows
1 negative ADR (-6.38)	Removed	87,229 rows
1 extreme ADR (5,400, cancelled)	Removed	87,228 rows
reservation_status_date stored as text	Converted to datetime	datetime64

Six new columns were then engineered: *total_stay_nights, total_guests, estimated_revenue, total_previous_bookings, booking_status, stay_length_category*. Final dataset: **87,228 rows x 38 columns**.

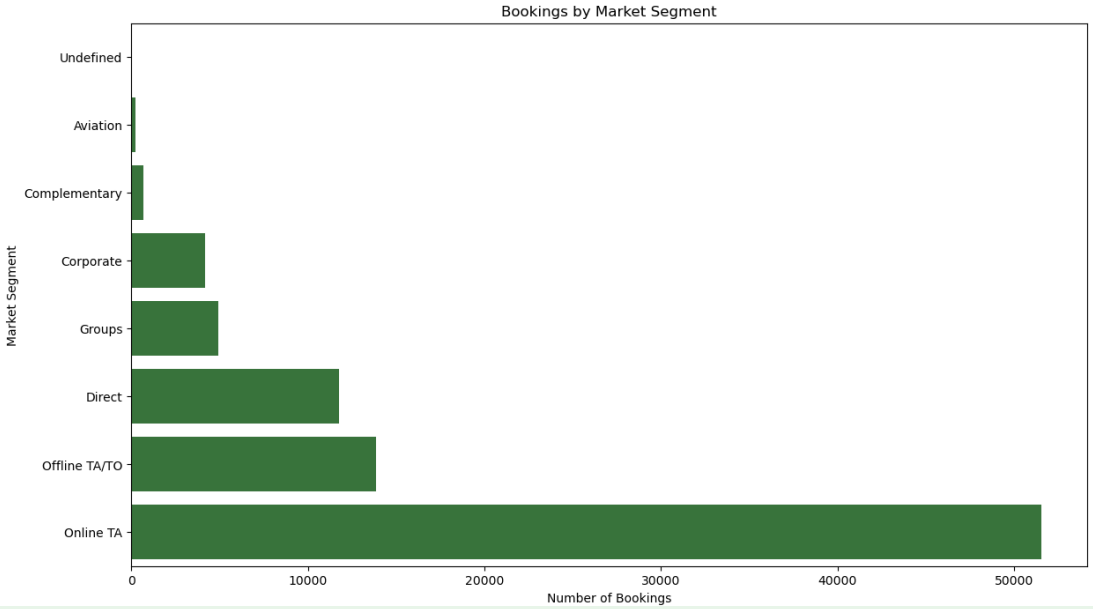
3. Key insights

3.1 City Hotel gets more bookings, but also more cancellations



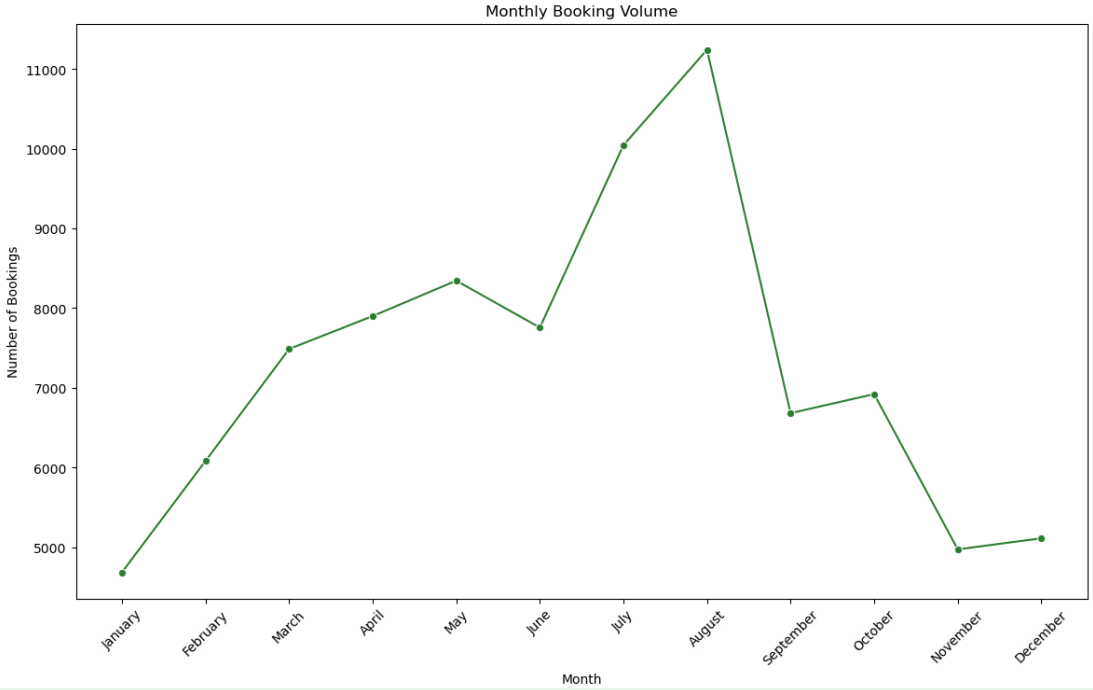
City Hotel received 53,273 bookings vs 33,955 for Resort Hotel, but 30.1% of City bookings were cancelled vs 23.5% at the Resort. City Hotel accounts for about 67% of all cancellations, so that is where cancellation control matters most.

3.2 Online travel agents drive volume and cancellations



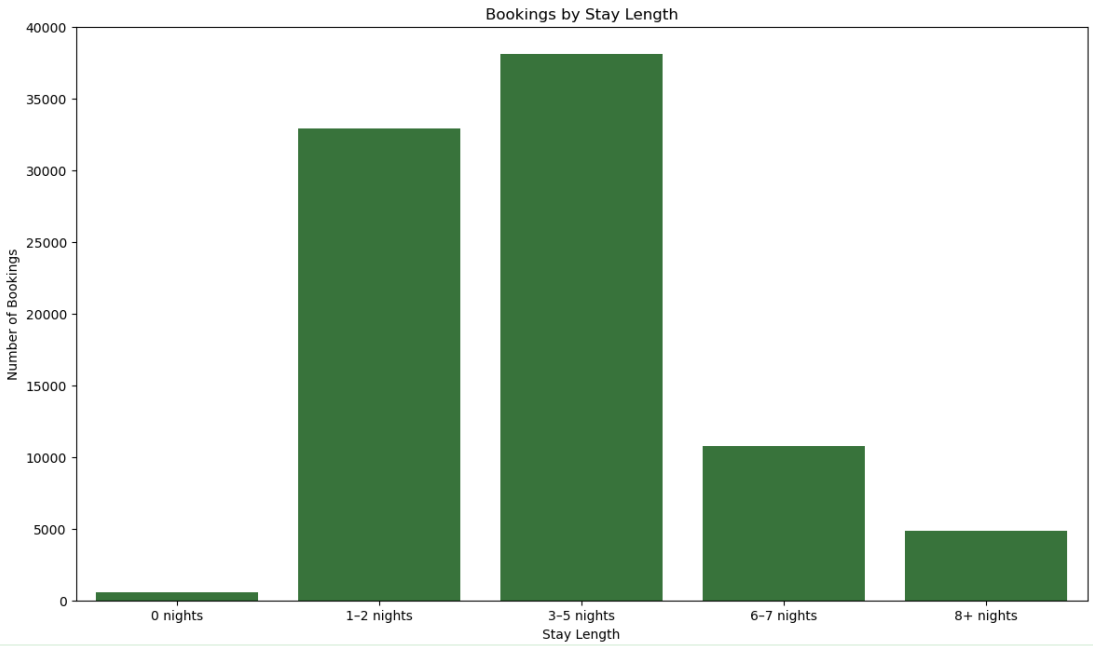
Online TA brought in 51,553 bookings (59% of the total) but 35.4% of them were cancelled, versus about 12 to 15% for Direct, Corporate and Offline TA/TO. That makes Online TA responsible for roughly 76% of all cancellations, which suggests the channel's flexible booking terms attract speculative reservations.

3.3 Demand peaks in summer



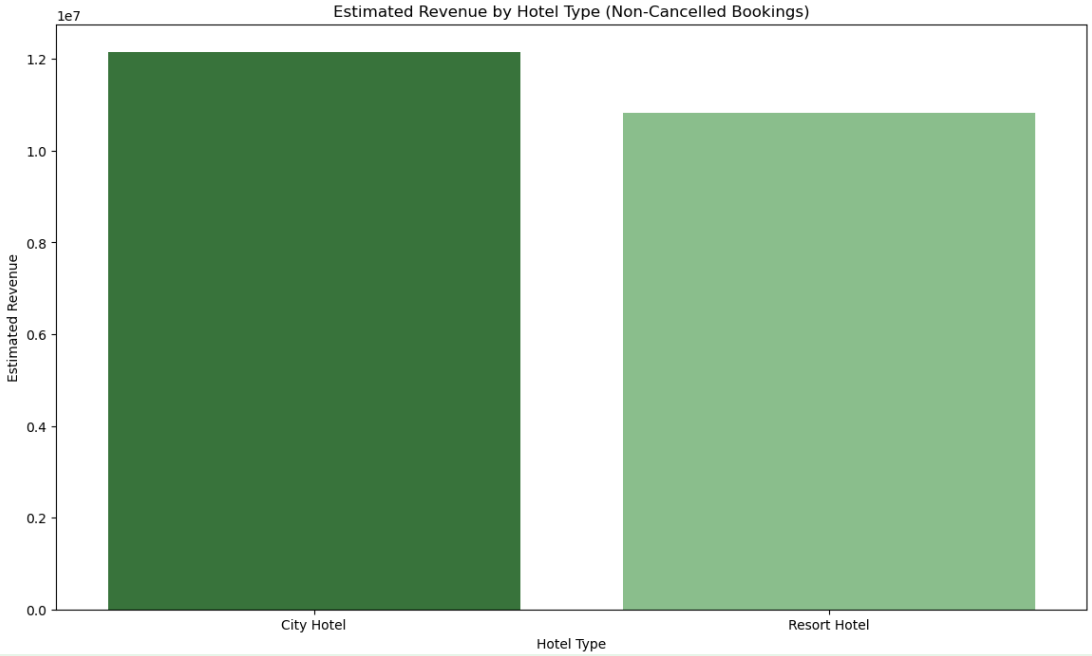
August (11,242 bookings) and July (10,043) are the busiest months, while January (4,685) is the quietest, a gap of 6,557 bookings. This points to leisure-driven seasonality, so staffing and pricing should follow the calendar.

3.4 Most guests stay 1 to 5 nights



71,023 bookings (81%) were for 1 to 5 nights, and the average stay was 3.6 nights. Short stays dominate, so room turnover efficiency matters more than long-stay packages.

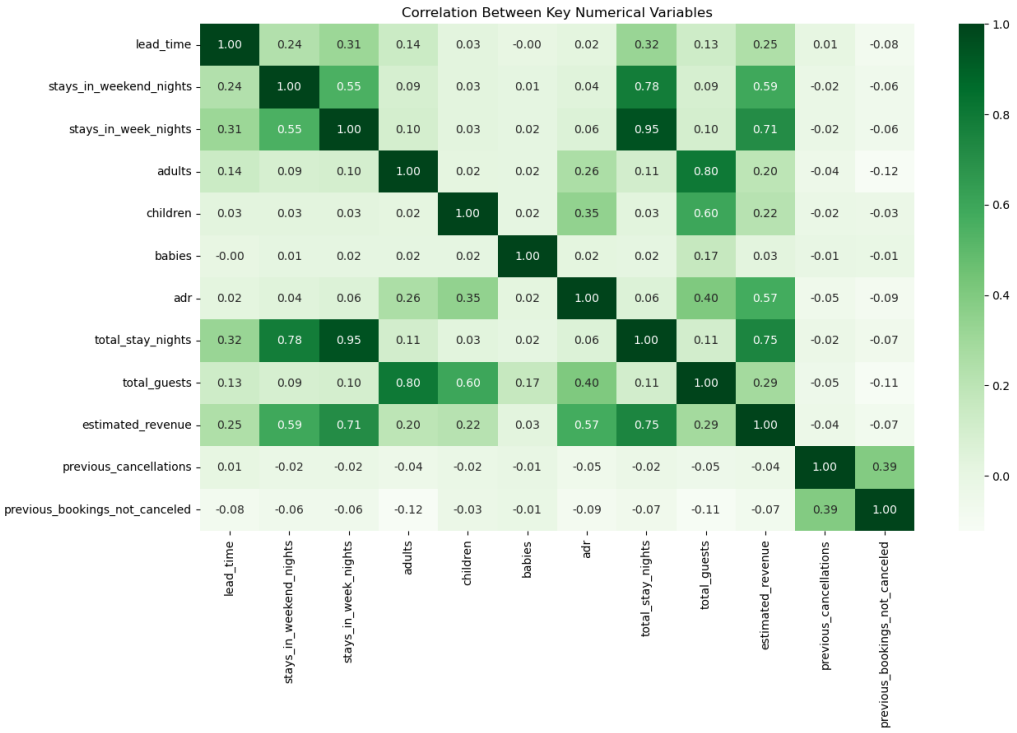
3.5 City Hotel earns more in total, but the Resort earns more per booking



Among non-cancelled bookings, City Hotel generated 12.15M in estimated revenue vs 10.82M for the Resort, but this is driven by volume: per booking the Resort earns 416.34 vs 326.29 for City Hotel (about 28% more). Resort guests either pay higher rates or stay longer, which is worth investigating.

Other findings

- **Overall cancellation rate:** 27.5% (24,008 of 87,228 bookings).
- **Lead time:** median 49 days, but ranging from same-day to 737 days.
- **Guests:** most bookings are for 2 guests; 82% are Transient (individual) customers.
- **Top source markets:** Portugal (27,354), UK (10,423), France (8,823).
- **Meals and deposits:** Bed and Breakfast is the most common meal plan (67,905); 98.7% of bookings have no deposit.
- **Room assignment:** Room type A was reserved 56,434 times but assigned only 46,282 times, so about 10,000 guests were placed in a different room type.
- **Correlation:** length of stay is strongly associated with estimated revenue, which is expected because revenue = ADR x nights.



4. Recommendations

Area	Recommendation
Online TA cancellations	Test stricter cancellation windows or partial deposits for Online TA bookings, and track cancellation rate by channel every month, not just volume.
City Hotel	Prioritise cancellation management here (30.1% rate). A controlled overbooking buffer based on past cancellation rates could protect occupancy.
Direct bookings	Direct and Corporate bookings cancel least (about 12 to 15%). Promote direct booking offers to shift demand towards more reliable channels.
Seasonal planning	Plan staffing, inventory and promotions around the July and August peak, and run off-season offers in January and November.
Resort Hotel	Resort bookings are worth more each. Grow Resort volume, especially in quieter months, to increase total revenue.
Revenue strategy	Review ADR and length of stay together by hotel, since revenue depends on both.

5. Limitations

- Estimated revenue is calculated as ADR x nights for non-cancelled bookings only. It is an estimate, not actual billed revenue, because extras, discounts and refunds are not in the data.
- The analysis shows association, not cause. For example, Online TA bookings cancel more, but the data cannot say why.

• Duplicate rows were treated as repeated records. In booking data, some identical rows could be genuine separate bookings.

• The data covers 2015 to 2017 and two hotels, so findings may not apply to other periods or properties.

6. Conclusion

After cleaning 119,390 raw records down to 87,228 reliable bookings, the analysis shows that **cancellations, not demand, are the main challenge**: more than a quarter of bookings are cancelled, and most of those come through Online TA and the City Hotel. Managing those two areas, planning around the summer peak and growing the higher-value Resort bookings would give the biggest gains.